YEAR 7 SCIENCE
EARTH AND SPACE SCIENCES
WORKSHEET - DAY/NIGHT AND SEASONS

Use Chapter 6.1 of Science Quest 7 to answer Q 1-10 given below.

UNDERSTANDING AND INQUIRING

REMEMBER

1 Explain the difference between the revolution and rotation of the Earth.

2 How long does it take the Earth to complete one:
(a) rotation
(b) revolution?

3 Explain why we experience day and night.

4 Why does the sun rise in the east and set in the west?

5 During which season does Australia tilt towards the sun?

6 Explain why it is usually warmer on a summer's day than on a winter's day.

7 Explain why there are 365 days in most years but 366 days in every fourth year.

THINK

8 Explain, with the aid of a diagram, why the South Pole is in darkness during the Southern Hemisphere's winter.

9 Use the diagram at right to answer the following questions.
(a) Which of the locations A, B, C, D and E:
(i) are in daylight
(ii) are experiencing summer
(iii) are experiencing the longest day
(iv) are experiencing the shortest day?
(b) In which of the locations that are in daylight will the sun set first?



10 Explain why both the time and position of sunrise and sunset are not the same every day.

BRAINSTORM

11 In a small group, brainstorm to compile a list of occupations in which day-to-day work is affected by seasonal changes. Provide a brief description of how each occupation is affected.

INVESTIGATE AND THINK

12 Observe the position of the shadow of a tree trunk or vertical pole from time to time on a sunny day.
(a) Explain how the shadow moves during the day from sunrise to sunset.
(b) How do you think prehistoric people have explained the movement of the shadow?
(c) What does the shadow tell you about the sun and the Earth?
(d) How would you expect the length of the shadow to change from summer to winter?
(e) In ancient times a vertical stick was used as a daylight clock. It was called a sundial. Describe the disadvantages of sundials.

INVESTIGATE

13 Use the internet or other resources to find out when and where the astronomer Al-Battani lived and worked, and what contribution he made to an understanding of the seasons.

14 Use the **Day, night and time zones** interactivity in your eBookPLUS to determine whether it is day or night in any location of the world at a specific time.
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